

# Assignment 3

Week 3

6733172621 Patthadon Phengpinij

*Collaborators.* Claude (for L<sup>A</sup>T<sub>E</sub>X styling and grammar checking)

## Problem 1. Finding Eigenvectors by Hand

This is a tedious exercise. But I want you to go through to practice the skill of finding eigenvectors. We can also measure the electron spin in  $\hat{y}$  direction by setting up an external magnetic field in the  $\hat{y}$  direction. It is given that the corresponding operator is

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Again, we are using the  $|0\rangle / |1\rangle$  basis. Find the eigenvalues and eigenvectors in the  $|0\rangle / |1\rangle$  basis.

**Solution.** To find the eigenvalues, we solve the characteristic equation  $\det(\sigma_y - \lambda I) = 0$ :

$$\det \begin{pmatrix} -\lambda & -i \\ i & -\lambda \end{pmatrix} = \lambda^2 - (-i)(i) = \lambda^2 - 1 = 0$$

which gives  $\lambda_1 = 1$  and  $\lambda_2 = -1$ .

**Eigenvector for  $\lambda_1 = 1$ .** Solve  $(\sigma_y - I) \vec{v} = \vec{0}$ :

$$\begin{pmatrix} -1 & -i \\ i & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

The first row gives  $-v_1 - iv_2 = 0$ , i.e.  $v_1 = -iv_2$ . Setting  $v_1 = 1$  gives  $v_2 = i$ , so the unnormalized eigenvector is  $\begin{pmatrix} 1 \\ i \end{pmatrix}$ . Normalizing (norm =  $\sqrt{1^2 + |i|^2} = \sqrt{2}$ ):

$$|\lambda_1 = 1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} = \frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle)$$

**Eigenvector for  $\lambda_2 = -1$ .** Solve  $(\sigma_y + I) \vec{v} = \vec{0}$ :

$$\begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

The first row gives  $v_1 - iv_2 = 0$ , i.e.  $v_1 = iv_2$ . Setting  $v_1 = 1$  gives  $v_2 = -i$ , so the unnormalized eigenvector is  $\begin{pmatrix} 1 \\ -i \end{pmatrix}$ . Normalizing:

$$|\lambda_2 = -1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} = \frac{1}{\sqrt{2}} (|0\rangle - i|1\rangle)$$

**Check.**

$$\sigma_y |\lambda_1 = 1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ i \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} = |\lambda_1 = 1\rangle$$

$$\sigma_y |\lambda_2 = -1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -i \end{pmatrix} = -\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} = -|\lambda_2 = -1\rangle$$

both of which confirm  $\sigma_y |\lambda_k\rangle = \lambda_k |\lambda_k\rangle$ , as expected.

**Problem 2. Finding Eigenvectors Using Python**

Use Google Colab to check your answer. What you need is the function to find the eigenvectors and eigenvalues. The following code first builds the matrix and then find the eigenvectors and eigenvalues:

```
1 sigmay = np.array([[0, -1j], [1j, 0]])
2 print(sigmay)
3
4 eigenvalues, eigenvectors = np.linalg.eig(sigmay)
5 print(eigenvalues)
6 print(eigenvectors)
```

**Solution.** Running the provided code in Google Colab prints

$$\text{eigenvalues} \approx \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \text{eigenvectors} \approx \begin{pmatrix} -0.7071i & 0.7071 \\ 0.7071 & -0.7071i \end{pmatrix}$$

where each column of the eigenvector matrix is the eigenvector for the eigenvalue in the corresponding position.

1. The first column,  $\begin{pmatrix} -0.7071i \\ 0.7071 \end{pmatrix} = -i \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$ , is the hand-calculated eigenvector for  $\lambda_1 = 1$  up to the global phase factor  $-i$ , which is physically irrelevant since an eigenvector is only defined up to a nonzero scalar multiple.
2. The second column,  $\begin{pmatrix} 0.7071 \\ -0.7071i \end{pmatrix} \approx \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$ , matches the hand-calculated eigenvector for  $\lambda_2 = -1$  exactly.

**Problem 3. Pauli Matrix Multiplication by Hand and Using Python**

Use hand calculation to show that  $\sigma_y^2 = I$ . Then verify with Colab using the following codes. Note that `np.matmul` is for matrix multiplication.

```
1 sigmay = np.array([[0, -1j], [1j, 0]])
2
3 print(sigmay)
4 print(np.matmul(sigmay, sigmay))
```

**Solution.** By hand:

$$\sigma_y^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} (-i)(i) & 0 \\ 0 & (i)(-i) \end{pmatrix} = \begin{pmatrix} -i^2 & 0 \\ 0 & -i^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

Verifying with the provided Python code, the output is

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma_y^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

which matches the hand-calculated result  $\sigma_y^2 = I$ .

**Problem 4. Pauli Matrix Multiplication by Hand and Using Python**

Repeat the previous question for  $\sigma_z^2 = I$  using both hand calculation and Colab.

**Solution.** By hand, with  $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ :

$$\sigma_z^2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1^2 & 0 \\ 0 & (-1)^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

Verifying with Colab:

```
1 sigmaz = np.array([[1, 0], [0, -1]])
2
3 print(sigmaz)
4 print(np.matmul(sigmaz, sigmaz))
```

which prints  $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$  and  $\sigma_z^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ , matching the hand-calculated result.

**Problem 5. Proof of Hermitianity**

Show that  $\sigma_y$  is Hermitian. Verify using Colab.

(*Hints:* find  $\sigma_y^{*T}$  and show that it is the same as  $\sigma_y$ )

**Solution.** By hand, the conjugate transpose of  $\sigma_y$  is obtained by transposing and then conjugating every entry:

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \implies \sigma_y^T = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix} \implies \sigma_y^{*T} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \sigma_y$$

Since  $\sigma_y^{*T} = \sigma_y$ ,  $\sigma_y$  is Hermitian.

Verifying with Colab:

```
1 sigmay_dagger = sigmay.conj().T
2
3 print(sigmay_dagger)
4 print(np.allclose(sigmay_dagger, sigmay))
```

which prints  $\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ , identical to  $\sigma_y$ , and True for the equality check, confirming  $\sigma_y$  is Hermitian.

**Problem 6. Operation of Paul Matrix on Bra and Ket**

Find  $\sigma_y |0\rangle$ ,  $\langle 0| \sigma_y$ . Are they the same? Why? Verify using Colab. You can use `np.dot` for matrix-vector multiplication. The following shows how to do  $\sigma_y |0\rangle$ .

```
1 zerospin = np.array([[1], [0]])
2
3 print(zerospin.shape)
4 print(np.dot(sigmay, zerospin))
```

**Solution.** First,  $\sigma_y |0\rangle$ :

$$\sigma_y |0\rangle = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ i \end{pmatrix} = i |1\rangle$$

Next,  $\langle 0| \sigma_y$ , where  $\langle 0| = (1 \ 0)$ :

$$\langle 0| \sigma_y = (1 \ 0) \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = (0 \ -i) = -i \langle 1|$$

They are **not** the same:  $\sigma_y |0\rangle = i |1\rangle$  while  $\langle 0| \sigma_y = -i \langle 1|$ , which differ by a sign on the imaginary coefficient. This is expected because, since  $\sigma_y$  is Hermitian (Problem 5),  $\langle 0| \sigma_y = (\sigma_y |0\rangle)^{*^T}$  is the *conjugate* transpose, not just the transpose, of  $\sigma_y |0\rangle$ ; as  $\sigma_y |0\rangle = i |1\rangle$  has a nonzero imaginary coefficient, conjugation flips its sign.

Verifying with Colab:

```
1 zerospin = np.array([[1], [0]])
2 print(zerospin.shape)
3
4 sigma_y_zero = np.dot(sigmay, zerospin)
5 print(sigma_y_zero)
6
7 bra_zero = zerospin.conj().T
8 bra_zero_sigma_y = np.dot(bra_zero, sigmay)
9 print(bra_zero_sigma_y)
```

which prints  $\begin{pmatrix} 0 \\ i \end{pmatrix}$  for  $\sigma_y |0\rangle$  and  $(0 \ -i)$  for  $\langle 0| \sigma_y$ , matching the hand-calculated result.

# APPENDIX 1

Assignment 03 | Code Solution

## assignment-03-code

August 28, 2026

# 1 Assignment 03

---

## 1.1 Import External Libraries

```
[1]: import numpy as np
import sympy as sp
from IPython.display import display, Math
```

---

## 1.2 2. Finding Eigenvectors Using Python

```
[2]: sigmay = np.array([[0, -1j], [1j, 0]])

display(Math(f"\\sigma_y = {sp.latex(sp.Matrix(sigmay))}"))

eigenvalues, eigenvectors = np.linalg.eig(sigmay)
eigenvalues = np.round(eigenvalues, 4)
eigenvectors = np.round(eigenvectors, 4)

display(Math(f"\\text{{Eigenvalues: }} {sp.latex(sp.Matrix(eigenvalues))}"))
display(Math(f"\\text{{Eigenvectors: }} {sp.latex(sp.Matrix(eigenvectors))}"))
```

$$\sigma_y = \begin{bmatrix} 0 & -1.0i \\ 1.0i & 0 \end{bmatrix}$$

$$\text{Eigenvalues: } \begin{bmatrix} 1.0 \\ -1.0 \end{bmatrix}$$

$$\text{Eigenvectors: } \begin{bmatrix} -0.7071i & 0.7071 \\ 0.7071 & -0.7071i \end{bmatrix}$$


---

### 1.3 3. Pauli Matrix Multiplication by Hand and Using Python

```
[3]: sigmay = np.array([[0, -1j], [1j, 0]])

display(Math(f"\\sigma_y = {sp.latex(sp.Matrix(sigmay))}"))
display(Math(f"\\sigma_y^2 = {sp.latex(sp.Matrix(np.matmul(sigmay, sigmay)))}"))
```

$$\sigma_y = \begin{bmatrix} 0 & -1.0i \\ 1.0i & 0 \end{bmatrix}$$

$$\sigma_y^2 = \begin{bmatrix} 1.0 & 0 \\ 0 & 1.0 \end{bmatrix}$$


---

### 1.4 4. Pauli Matrix Multiplication by Hand and Using Python

```
[4]: sigmaz = np.array([[1, 0], [0, -1]])

display(Math(f"\\sigma_z = {sp.latex(sp.Matrix(sigmaz))}"))
display(Math(f"\\sigma_z^2 = {sp.latex(sp.Matrix(np.matmul(sigmaz, sigmaz)))}"))
```

$$\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\sigma_z^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$


---

### 1.5 5. Proof of Hermitianity

```
[5]: sigmay_dagger = sigmay.conj().T

display(Math(f"\\sigma_y^{\\{\\ast\\}T\\}} = {sp.latex(sp.Matrix(sigmay_dagger))}"))
print(f"Check if Sigma Y is Hermitian: {np.allclose(sigmay_dagger, sigmay)}")
```

$$\sigma_y^{*T} = \begin{bmatrix} 0 & -1.0i \\ 1.0i & 0 \end{bmatrix}$$

Check if Sigma Y is Hermitian: True

---

### 1.6 6. Operation of Paul Matrix on Bra and Ket

```
[6]: zerospin = np.array([[1], [0]])
print("Dimension of zerospin:\n", zerospin.shape)

sigma_y_zero = np.dot(sigmay, zerospin)
display(Math(f"\\sigma_y \\vert 0 \\rangle = {sp.latex(sp.
    Matrix(sigma_y_zero))}"))
```

```
bra_zero = zerospin.conj().T
bra_zero_sigma_y = np.dot(bra_zero, sigmay)
display(Math(f"\\langle 0 \\vert \\sigma_y = {sp.latex(sp.
    ↵Matrix(bra_zero_sigma_y))}"))
```

Dimension of zerospin:

(2, 1)

$$\sigma_y|0\rangle = \begin{bmatrix} 0 \\ 1.0i \end{bmatrix}$$

$$\langle 0|\sigma_y = [0 \quad -1.0i]$$


---